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# Comparative Analysis of Machine Learning Algorithms for Early Detection of Lung Cancer

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**Abstract.** Lung cancer continues to be a major global health concern, contributing significantly to cancer-related mortality. Late stage diagnosis of lung cancer remains a critical challenge, often leading to reduced survival rates. This study explores the potential of various machine learning classification algorithms, including K-Nearest Neighbors (KNN), Naive Bayes (NB), Decision Tree (DT), Random Forest (RF) and Support Vector Machine (SVM), for early and accurate detection and diagnosis of lung cancer. A comprehensive dataset was used to evaluate the performance of the machine learning models based on key performance metrics such as accuracy, precision, recall, F1 score, and ROC-AUC. Among the models analyzed, the Support Vector Machine model demonstrated superior classification performance, achieving the highest ROC-AUC value of 0.99793. Training accuracy values of the machine learning models ranged from 87.44% (Naive Bayes) to 97.32% (K-Nearest Neighbors, Random Forest and Decision Tree), while Support Vector Machine model training accuracy is about 95.88% and all five machine learning models consistently exhibited a testing accuracy of about 98.14%. The study also highlights the robustness of ensemble methods, such as Random Forest, and the discriminative capabilities of Support Vector Machine for early-stage lung cancer detection. The findings emphasize the integration of various machine learning models into clinical diagnostics to improve the efficiency, reduce dependency on invasive procedures, and also helps to improve patient outcomes.

**Keywords:** Lung Cancer Detection, K-Nearest Neighbors (KNN), Naive Bayes (NB), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), ROC-AUC analysis.

## 1 Introduction

Lung cancer, often labeled a silent killer, continues to pose a critical global health problem, representing a significant proportion of cancer-related deaths worldwide. Despite advances in treatment methodologies, late stage diagnosis remains a substantial factor contributing to high mortality rates. Recent studies indicate

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that early and precise detection of lung cancer significantly improves patient survival by facilitating timely medical interventions [1,2]. However, conventional diagnostic techniques, such as biopsies and radiological imaging, often involve invasive procedures, high costs, and susceptibility to human error, highlighting the need for more efficient and non-invasive detection methodologies [3,4].

Machine learning (ML) has significantly transformed medical diagnostics by enabling efficient analysis of complex datasets with minimal human intervention. Compared to traditional diagnostic methods, machine learning based solutions offer reliable predictions while reducing the reliance on invasive procedures [5, 6]. Classification algorithms such as Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Decision Tree (DT), and Naive Bayes (NB) have shown promising results in lung cancer detection through effective analysis of radiological images and patient data [7,8]. Furthermore, ensemble methods, including RF, have gained recognition for robustness in classification tasks, further enhancing predictive precision [9,10]. Clinical implementations of artificial intelligence, as described in [11], have further highlighted the potential to revolutionize early detection methodologies and reduce diagnostic delays.

Several studies have demonstrated the efficacy of machine learning algorithms in the detection of lung cancer. Deep learning techniques, explored in [7], have shown a high precision in analyzing radiological data such as CT scans. Similarly, in [8], a systematic review of deep learning algorithms is presented, revealing superior sensitivity and specificity in the diagnosis of lung cancer. Recent reviews emphasize the potential of ensemble methods, such as RF, to achieve robust classification outcomes [9,10]. Despite these advances, challenges persist in the practical application of machine learning for the detection of lung cancer.

Numerous studies have concentrated mainly on individual algorithms or specific datasets, often lacking comprehensive comparative analyses to determine the most effective model [12,13]. Moreover, issues such as the variability of the dataset, the class imbalance, and the feature selection methodologies impact the generalizability of these models, thereby limiting their applicability in the real world [14,15]. Addressing these limitations requires a systematic evaluation of various machine learning algorithms in diverse datasets to develop a robust predictive framework [16,17].

## 1.1 Motivation

The persistent challenge of high mortality rates due to late-stage lung cancer diagnosis underscores the need for early, accurate, and noninvasive detection techniques. The limitations inherent in traditional diagnostic methods, such as invasiveness, high cost, and susceptibility to human error, motivate the exploration of advanced machine learning based diagnostic approaches. Furthermore, disparities in healthcare access and outcomes, emphasize the urgency of customized machine learning solutions capable of addressing diverse populations effectively.

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## 1.2 Contribution

This study contributes by performing a comparative analysis of five widely used machine learning algorithms which are: KNN, NB, DT, RF, and SVM has been used for identifying the most effective algorithm for lung cancer detection. The machine learning models were evaluated based on key performance metrics, such as accuracy, precision, recall, F1 score, and ROC-AUC. By determining the optimal machine learning algorithm, this research aims to facilitate the integration of artificial intelligence into clinical diagnostics, minimize reliance on invasive procedures, and significantly improve patient outcomes.

## 2 Background

In recent years, the role of machine learning (ML) in the early detection of lung cancer has been extensively explored, using various medical data sources. In this section, a comprehensive overview of significant research contributions in this domain is provided.

In paper [3], the application of machine learning techniques such as Support Vector Machine (SVM), Bacterial Foraging Optimization (BFO), Linear Discriminant Analysis (LDA), and neural networks was explored for lung cancer detection, highlighting their capability to improve early diagnosis and patient survival rates. Similarly, an alternative approach presented in [5] compared several classification methods, including Naive Bayes, SVM, Decision Tree, and Logistic Regression. Their findings highlighted that Logistic Regression model achieved the highest accuracy (96.9%) on the UCI dataset, while SVM model delivered superior performance of accuracy (99.2%) on an alternative dataset from data.world. Further studies such as [4, 6] also emphasized the importance of feature selection and deep learning architectures, demonstrating high precision rates of up to 99% using CNN for effective segmentation and feature extraction from medical images.

An extensive review of classification methodologies for lung cancer detection was presented in [10], comparing various machine learning models including CNN, SVM, ANN, MLP, KNN, RF and EDM. CNN was reported to have the highest accuracy (96%) with small datasets, while EDM showed a lower accuracy (77.8%). Furthermore, studies such as [7, 8, 12] provided significant insights into deep learning techniques, including CNN and transfer learning, demonstrating their effectiveness and robustness to classify lung cancer from radiological images. In addition, the evaluation of the ensemble methods in lung cancer detection was highlighted in [9, 13], highlighting their precision and reliability for real-world clinical applications.

Recent research has also investigated the potential of advanced deep learning architectures, including Generative Adversarial Networks (GAN) and Transformer-based networks, to improve diagnostic performance by augmenting data sets and fine-tuning models on medical imaging data [16–18]. In addition, studies such as [19–22] explored deep feature learning techniques, confirming that automated

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segmentation and deep feature extraction significantly improve the accuracy and reliability of machine learning models in the diagnosis of lung cancer. Such comprehensive evaluations underscore the transformative potential of integrating advanced machine learning and deep learning methodologies into healthcare care, enabling improved detection accuracy and patient outcomes.

As summarized in Table 1, various machine learning models have been explored for the detection of lung cancer.

**Table 1.** Summary of Literature Survey

| Study | Model                                                | Dataset                        | Accuracy (%)                     |
|-------|------------------------------------------------------|--------------------------------|----------------------------------|
| [3]   | SVM, BFO, LDA, Neural Net-works                      | Healthcare data                | N/A                              |
| [5]   | Naïve Bayes, SVM, Decision Tree, Logistic Regression | UCI ML Repository, data.world  | 96.9 (LR), 99.2 (SVM)            |
| [4]   | Naïve Bayes, SVM, ANN, Logistic Regression           | Lo- Biomedical Images          | N/A                              |
| [6]   | ANN, CNN, SVM, GLCM, DWT                             | CT Images                      | CNN: 99, ANN: 97, SVM: 96        |
| [10]  | CNN, SVM, ANN, MLP, KNN, RF, EDM                     | CT Scan Images                 | CNN: 96, EDM: 77.8               |
| [7]   | CNN, Transfer Learning                               | X-ray, CT Scan                 | N/A                              |
| [13]  | K-NN, RF, SVM                                        | Lung Cancer Dataset            | RF: 84.2, SVM: 82.1              |
| [8]   | Deep Learning Meta-Analysis                          | Multiple Studies               | Sensitivity: 93, Specificity: 68 |
| [12]  | DNN, K-NN, SVM, GMM                                  | Radiological Images            | N/A                              |
| [14]  | Deep Learning CNN                                    | Large-scale Medical Images     | N/A                              |
| [1]   | ML Risk Prediction Models                            | Clinical Data                  | High Sensitivity, Specificity    |
| [2]   | ML Models for Prediction                             | Clinical and Radiological Data | N/A                              |
| [19]  | Deep Feature Learning                                | Medical Image Analysis         | N/A                              |
| [18]  | GAN-based Augmentation                               | Laparoscopic Videos            | Improved Accuracy                |
| [16]  | Transfer Learning CNN                                | PET/CT Scan Data               | 91.3                             |
| [17]  | Deep Neural Networks                                 | Brain Tumor Segmentation       | 92.1                             |
| [21]  | Deep Feature Extraction                              | Tumor Classification           | N/A                              |
| [22]  | Fine-tuned CNN                                       | MRI Data                       | 89.7                             |
| [23]  | CNN for Liver Tumor Detection                        | CT Images                      | 88.5                             |

### 3 Machine Learning Algorithms Used in Lung Cancer Detection

Various machine learning (ML) algorithms have been extensively studied for lung cancer detection, each demonstrating unique advantages in precision, interpretability, and efficiency. The key algorithms applied to patient records, ra-

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diological images, and clinical datasets include K-Nearest Neighbors (KNN), Naive Bayes (NB), Decision Tree (DT), Random Forest (RF) and Support Vector Machine (SVM). Early detection facilitated by these algorithms significantly improves patient survival rates [3–6].

- **K-Nearest Neighbors (KNN)** classifies new data based on the majority class among nearest neighbors. It shows moderate precision in classifying tumor characteristics, but requires optimization for large datasets [13].
- **Naive Bayes (NB)** employs probabilistic classification based on Bayes' Theorem. It efficiently handles categorical and numerical data due to the assumption of feature independence. However, this assumption may limit the accuracy with complex medical datasets [2].
- **Decision Tree (DT)** provides high interpretability using a tree structure based on feature values. Although effective in predicting malignancy using tumor characteristics and patient demographics, DT often suffers from the overfitting of high-dimensional medical data [1].
- **Random Forest (RF)**, an ensemble method, enhances decision trees by aggregating predictions from multiple trees. It significantly reduces overfitting and achieves high accuracy, especially in distinguishing benign and malignant lung nodules from CT scans [9, 10].
- **Support Vector Machine (SVM)** excels with high-dimensional data, classifying patients by identifying an optimal separation hyperplane. SVM demonstrates superior precision compared to other methods, particularly in differentiating malignant from benign tumors [7, 8].
- **Deep Learning techniques** such as convolutional neural networks (CNN) and transfer learning methods further improve lung cancer detection, particularly through automated feature extraction from radiological images, showing higher accuracy than traditional methods [12, 14].

Despite advances, challenges such as dataset variability, class imbalance, and explainability persist. Future research may focus on hybrid models that integrate traditional ML with deep learning techniques to improve accuracy and minimize diagnostic errors [18–20].

## 4 Results

This section presents the results obtained through the evaluation of various machine learning models using the Kaggle lung cancer dataset [24]. The performance of these machine learning models was evaluated based on performance metrics such as accuracy, precision, recall, F1 score, and ROC-AUC. A detailed analysis of the performance of the machine learning model was performed by presenting confusion matrices, ROC curves, and comparison of both the training and testing accuracy.

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#### 4.1 Dataset Description

The dataset used in this study was obtained from Kaggle [24]. It comprises key features relevant to the diagnosis of lung cancer, including various clinical and demographic characteristics. Preprocessing of data was performed in order to normalize numerical characteristics and encode categorical variables, ensuring the suitability of the dataset for subsequent analysis. Synthetic Minority Over-sampling Technique (SMOTE) has been used to convert the imbalanced dataset into a balanced dataset. Feature Selection was performed for the purpose of increasing the accuracy of the different machine learning models and also for selecting the most important features in the dataset.

#### 4.2 Performance Metrics

The following metrics were used for the evaluation of the models:

- **Training Accuracy:** The extent to which patterns were learned from the training dataset.
- **Testing Accuracy:** The ability of the model to generalize when applied to unseen data.
- **Precision, Recall, and F1-Score:** Metrics through which classification performance was assessed, accounting for both false positives and false negatives.
- **ROC-AUC:** The Area Under the Receiver Operating Characteristic curve, which represented the model's effectiveness in distinguishing between classes.

#### 4.3 Model-Wise Analysis with Results

The performance of each machine learning model was examined in detail based on the value of accuracy of the training and the testing, along with evaluation metrics such as precision, recall, F1 score, and ROC-AUC. These metrics enabled a comprehensive assessment of the effectiveness of the models in differentiating between positive and negative lung cancer cases. Visual analysis was carried out using confusion matrices and ROC curves to better understand classification performance and the balance between sensitivity and specificity. A summary of the findings for each model is presented below.

The confusion matrices provided insight into the reliability of each machine learning model by displaying the number of true positives, true negatives, false positives, and false negatives. Similarly, the ROC curves visualized the discriminative power of the models across varying thresholds. The performance of each machine learning model is discussed individually, with an emphasis on its strengths and limitations in lung cancer detection.

**Decision Tree (DT)** The decision tree model achieved a training accuracy of 97.32% and a testing accuracy of 98.14%. The precision, recall and F1 score were recorded as 0.96666, 1.0, and 0.98305, respectively. Strong classification capability was reflected in the ROC-AUC score of 0.97724.

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**K-Nearest Neighbors (KNN)** For the KNN model, training and testing accuracies of 97.32% and 98.14% were obtained. Precision and recall were measured at 0.96666 and 1.0, respectively, resulting in an F1 score of 0.98305. A ROC-AUC score of 0.97448 further supported its strong performance.

**Naive Bayes (NB)** The Naive Bayes classifier achieved a training accuracy of 87.44%, which was lower than other models, while the test accuracy remained at 98.14%. Precision and recall were measured at 0.96666 and 1.0, yielding an F1 score of 0.98305. A robust classification capability was confirmed by a ROC-AUC score of 0.98827.

**Random Forest (RF)** Training and testing accuracies of 97.32% and 98.14% were recorded for the Random Forest model. The precision, recall and F1 score remained consistent at 0.96666, 1.0, and 0.98305, respectively. Its reliability in classification was demonstrated by a ROC-AUC score of 0.98206.

**Support Vector Machine (SVM)** The Support Vector Machine model achieved a training accuracy of 95.88% and a testing accuracy of 98.14%. The precision, recall and F1 score were measured at 0.96666, 1.0, and 0.98305, respectively. The highest ROC-AUC score among all models, 0.99793, confirmed the superior performance of SVM in lung cancer classification.

### 4.4 Performance Comparison of All Machine Learning Models

The training and testing accuracy of each machine learning model was analyzed to gain valuable insights into their performance. Training accuracy was used to assess how effectively patterns had been learned from the training dataset, while testing accuracy reflected the ability of the models to generalize to unseen data. When consistent performance was observed across both metrics, it was inferred that neither underfitting or overfitting had occurred. In addition, the classification capabilities of the machine learning models were further evaluated through their ROC curves. These curves, which illustrate the trade-off between sensitivity and specificity, allowed the identification of the most effective models for class discrimination. Such analyses were deemed essential for selecting optimal models in real-world applications.

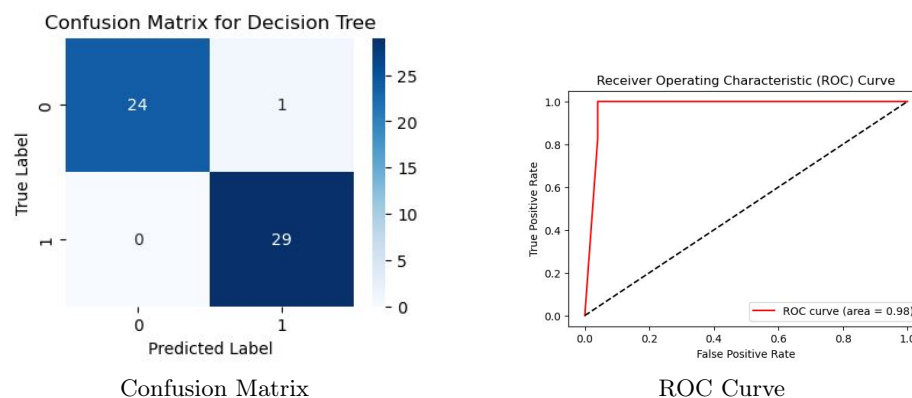
### 4.5 Visualization of Results

In this section, visual representations of the results are provided, including confusion matrices, ROC curves, and accuracy comparisons for all models. The confusion matrices were used to highlight the classification outcomes for each model in terms of true positives, true negatives, false positives, and false negatives. ROC curves were plotted to visualize the trade-off between sensitivity and

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specificity. In addition, a combined ROC curve and an accuracy comparison plot were generated to offer a comprehensive overview of the models' performances.

The classification results and the trade-off between sensitivity and specificity for each machine learning model used in this study are illustrated in Figures 1, 2, 3, 4, and 5. For each model, Figure (a) displays the confusion matrix, providing detailed information on correctly and incorrectly classified instances, while Subfigure (b) presents the ROC curve, which illustrates the model's ability to differentiate between classes. The Decision Tree model (Figure 1) is shown to have delivered robust classification results with a clear trade-off between sensitivity and specificity. Similarly, the models K-Nearest Neighbors (Figure 2) and Naive Bayes (Figure 3) exhibited strong performance, as reflected in their respective confusion matrices and ROC curves. The Random Forest model (Figure 4) was found to perform with high precision and strong classification capability. Lastly, the Support Vector Machine (SVM) model (Figure 5) was shown to provide superior classification performance, maintaining a well-balanced trade-off between sensitivity and specificity, as supported by its confusion matrix and the ROC curve. Together, these visualizations offer a detailed comparison of the classification performance of all evaluated models.



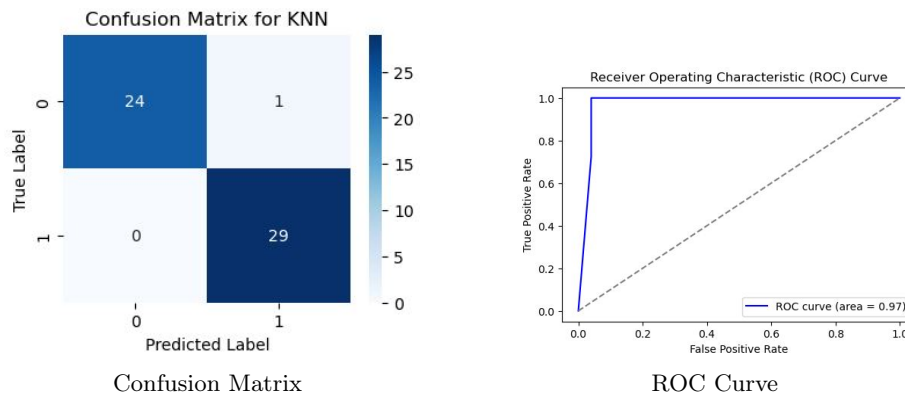
**Fig. 1.** Performance of the Decision Tree Model. (a) Confusion Matrix shows classification outcomes. (b) ROC Curve represents the trade-off between sensitivity and specificity.

The comparison of accuracy for all the machine learning models evaluated is shown in Figures 6(a) and 6(b). Figure 6(a) illustrates the testing accuracy of all models, reflecting their ability to effectively generalize to unseen data. Identical testing accuracy values were observed across all models, indicating consistent and reliable performance during evaluation.

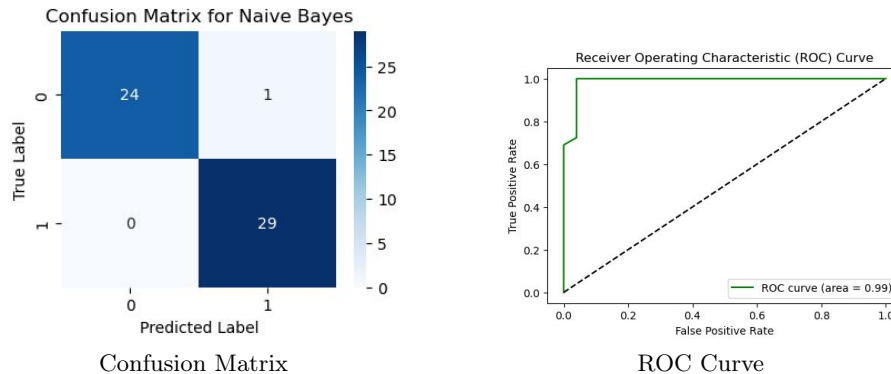
Figure 6(b) presents the training accuracy of the models, providing insight into how well each algorithm learned from the training dataset. Although most models achieved near-perfect accuracy during training, a marginally lower value

## Performance Analysis of ML Algorithms for Detection of Lung Cancer

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**Fig. 2.** Performance of the KNN Model. (a) Confusion Matrix shows classification outcomes. (b) ROC Curve represents the trade-off between sensitivity and specificity.



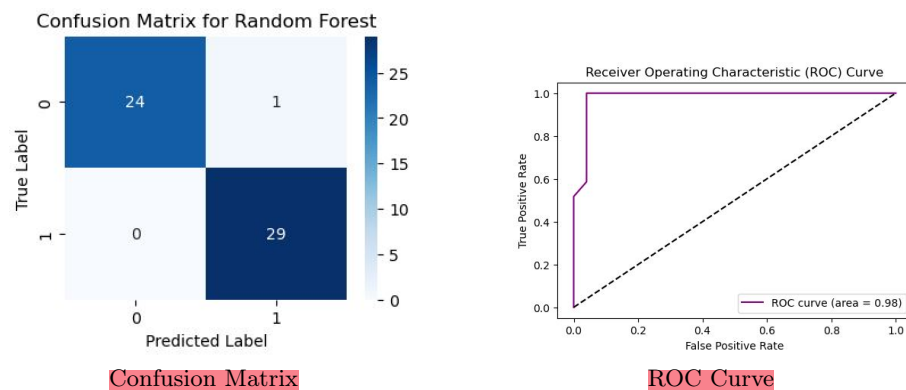
**Fig. 3.** Performance of the Naive Bayes Model. (a) Confusion Matrix shows classification outcomes. (b) ROC Curve represents the trade-off between sensitivity and specificity.

was observed for the Nave Bayes model, attributed to its inherently simplistic probabilistic framework.

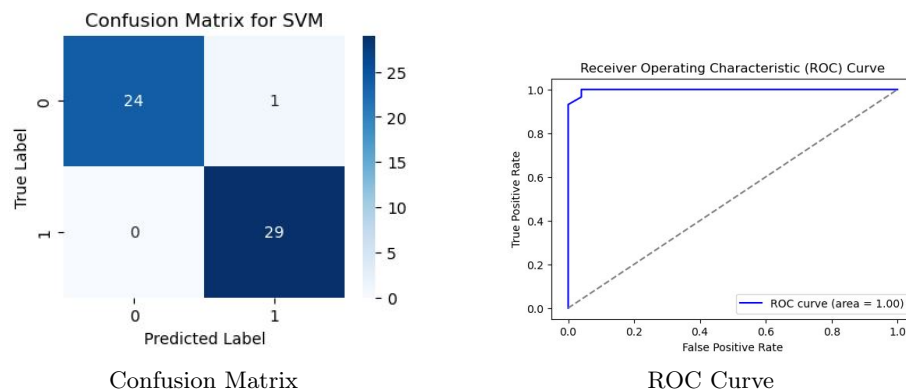
## 5 Conclusion and Future Scope

The effectiveness of machine learning algorithms in the early detection of lung cancer has been demonstrated through rigorous evaluation on structured datasets. Classification models which include Support Vector Machine (SVM), Random Forest, Decision Tree, K-Nearest Neighbors (KNN) and Naive Bayes were evaluated using various performance metrics such as accuracy, precision, recall, F1 score and ROC-AUC. Among these, machine learning models, Support Vector Machine exhibited the highest discriminative capacity. High testing accuracy was

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**Fig. 4.** Performance of the Random Forest Model. (a) Confusion Matrix shows classification outcomes. (b) ROC Curve represents the trade-off between sensitivity and specificity.

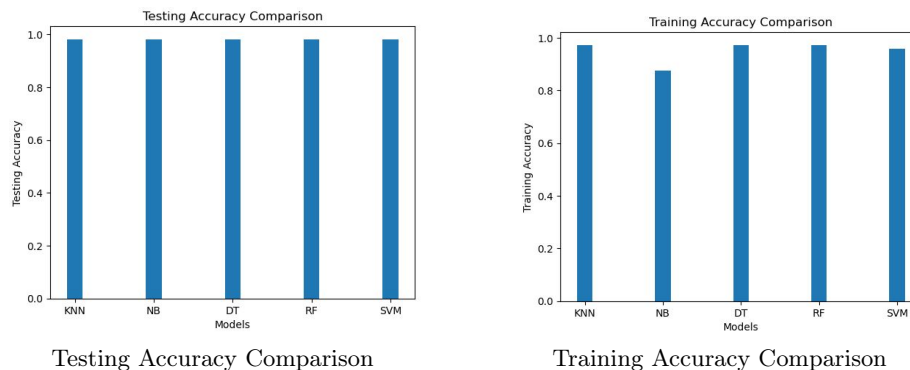


**Fig. 5.** Performance of the SVM Model. (a) Confusion Matrix shows classification outcomes. (b) ROC Curve represents the trade-off between sensitivity and specificity.

consistently observed across all machine learning models, and the ROC curves facilitated a comprehensive analysis of the sensitivity-specificity trade-offs. These results highlight the potential of machine learning algorithms to improve diagnostic accuracy and reduce the reliance on invasive procedures.

For future research, the incorporation of deep learning models such as Convolutional Neural Networks (CNNs) and ensemble learning strategies are recommended to improve robustness and scalability. The use of larger and more heterogeneous datasets is encouraged to address the challenges of generalization and class imbalance. In addition, the application of better feature selection method and dimensionality reduction techniques may further optimize the efficiency of the machine learning model. The adoption of Explainable AI (XAI) frameworks should also be explored to enhance model transparency and also clinical trust.

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**Fig. 6.** Accuracy Comparison for Machine Learning Models. (a) Testing Accuracy indicates generalization performance. (b) Training Accuracy reflects the learning effectiveness of each model.

These advances can support the real-time deployment of intelligent diagnostic systems in healthcare settings.

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